

Piyush Bhuyan

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Summary

Applied AI Engineer with 1.5+ years building production AI systems, machine learning pipelines, and scalable backend services. Building autonomous incident orchestrators across LLM reasoning, structured function calling, deterministic rule engines, and AWS serverless infrastructure. Focused on multi-agent systems, distributed pipelines, RAG, observability, and cloud-native architectures.

Work Experience

Keyloop India | Hyderabad

Applied AI Engineer

Jul 2026 – Present

- Architecting a serverless, autonomous incident orchestrator spanning **38 enterprise products** with a **6-tier deterministic-AI cascade**, integrating function calling, multi-agent orchestration, and BM25 pattern matching, routing **80% of events at \$0 LLM cost** via deterministic rules, targeting **75%+ zero-touch autonomous resolution** at a blended **\$0.03/event AI cost**.
- Reduced P2 MTTR by **99%** (90 min to under 90 sec) and major-incident resolution time by **97%** (5h 53m to under 10 min) by correlating real-time signals, change activity, and cross-team case data through circuit-breaker-guarded parallel enrichment from 6 data sources, with autonomous remediation via Paramiko-based execution agents.

Keyloop India | Hyderabad

Associate Engineer, AI Enablement

Jun 2025 – Jul 2026

- Designed and shipped a full-stack AI operations platform (FastAPI + React/TypeScript + MySQL + Docker) with **40+ REST API endpoints** and Grafana/Prometheus observability, achieving **40x throughput improvement** (5 to 200 req/s) via thread-safe caching, MySQL connection pooling, and GZIP compression reducing payload size **90%**, scaling to **500+ concurrent engineers**.
- Eliminated an N+1 query bottleneck by replacing 31 sequential database calls with batch IN-clause queries and a thread-safe connection pool, reducing critical-path latency **620x** and enabling **50x user capacity growth** (10 to 500 concurrent users).
- Built a hybrid generative AI/NLP triaging pipeline integrating Claude Sonnet with TF-IDF + cosine similarity for infrastructure issue classification (CPU/Memory/Disk/Swap) and cascade alert pattern detection, cutting **shift handover generation time 93%** (30–45 min to 2–3 min) and saving **30+ engineering hours/month**.

Skills

Languages & Backend: Python, Java, SQL, TypeScript/JavaScript, React, FastAPI, REST APIs, MySQL, Git

LLM & Agentic AI: Generative AI, Anthropic Claude, AWS Bedrock, Function Calling, RAG, Multi-agent Systems, BM25, TF-IDF, KMeans, LangChain, LangGraph, LangSmith, Embeddings, Vector Databases

Cloud & Infrastructure: AWS (Lambda, DynamoDB, SQS, S3, Bedrock), Docker, Kubernetes, Nginx, GitHub Actions, CI/CD

Machine Learning: Scikit-learn, XGBoost, LightGBM, PyTorch, HuggingFace Transformers, NLP, Pandas, NumPy, MLflow, DVC

Core Competencies: System Design, Distributed Systems, Agentic Workflows, Agile, Data Structures & Algorithms

Projects & Publications

pispyre | *Python, Static Analysis, AST, CLI Development, SQLite, Ruff, MyPy, Pytest* | [Repo]

2026

- Built **pispyre**, an open-source static analysis CLI that detects hallucinated imports in AI-generated Python code before runtime, deployable as a **CI gate, Python library, and pre-commit hook**.
- Engineered a modular rule engine implementing six analyses for phantom packages, removed stdlib modules, invalid attributes, abandoned dependencies, and moved symbols, with JSON and SARIF reporting for automated code quality workflows.

Layered Security - Gradient Boosting Meets Naive Bayes for Intrusion Detection | Springer CIS 2025 | [Link]

- Developed a double-layered intrusion detection system combining **Naive Bayes, XGBoost, and GAN-based augmentation**, improving minority-class detection by **14.49pp for R2L** and **5.42pp for U2R** on NSL-KDD.
- Replaced SVM with **XGBoost** and augmented rare attack classes using GANs, achieving an **80.20% F1-score** with reported gains of **4.5% in recall** and **5.4% in false-alarm rate** over the SVM-based second layer.

Education

Vasavi College of Engineering, Hyderabad

2021 – 2025

• *Bachelor of Engineering, Information Technology*

CGPA: 9.04/10

Certifications

AWS Certified Cloud Practitioner (CLF-C02) | AWS

2026

Certified SAgE 6 Practitioner | SAgE by Scaled Agile

2026